OFFICIAL COMMERCIAL & TECHNICAL PROPOSAL

QUOTATION FOR 5 KW DCR Bifacial Mono PERC ON-GRID SOLAR PLANT

High-Efficiency Rooftop Photovoltaic Power Generation System with Net Metering

QUOTATION REF	DATE OF ISSUE	VALIDITY PERIOD	SYSTEM CAPACITY
RSP/QTN/2026-27/0010	2026-09-21	15 Days (2026-10-06)	5 kWp Grid-Tied

CLIENT DETAILS & INSTALLATION SITE

Shri Rameshwar Patel
Patel Agro Industries
Phone: +91 98795 44321
Email: rameshwar.patel@example.com
Site: Survey No. 142/2, Near Industrial Area, Mandir Hasaud
City: Raipur, Chhattisgarh
GSTIN: 22AAAAA0000A1Z5

EPC CONTRACTOR & SERVICE PROVIDER

REJOY SOLAR POWER PRIVATE LIMITED
CREDA Approved Solar EPC Contractor
Plot No. 14, St No 14/b, Pragati Nagar Risali
Civic Centre, Bhilai, Durg, CG - 490006
Phone: +91 9770577527 / +91 7803844457
Email: rejoysolarpower@gmail.com
Web: www.rejoysolarpower.in | CIN: U40106CT2021PTC011234

EXECUTIVE PROJECT SUMMARY

Parameter	Technical & Commercial Specification
System Capacity & Topology	5 kWp Rooftop Grid-Tied (On-Grid) Photovoltaic Plant
Solar PV Modules	Waaree 545W DCR Bifacial Dual Glass Mono PERC Modules
Solar Grid Inverter	Waaree 5KW 3PH Grid Tie High Efficiency String Inverter with WiFi
Rooftop Mounting Structure	Elevated HDG / GI Structure
Estimated Daily Power Generation	~21.0 Units (kWh) per day (Weather & Irradiance dependent)
Estimated Annual Generation	~7,300 Units (kWh) per year
Government Subsidy Support	Eligible under PM Surya Ghar Muft Bijli Yojana (₹1,08,000)
Net Customer Project Investment	₹2,12,000 (Turnkey supply, installation & commissioning)

CREDA APPROVED
Govt. of Chhattisgarh Agency

ISO 9001:2015 CERTIFIED
Quality Management Standards

PM SURYA GHAR YOJANA
National Portal Subsidized EPC

Rejoy Solar Power

1. KEY BENEFITS OF SOLAR ON-GRID SYSTEM

Zero Monthly Peak Cost

Drastic Electricity Bill Reduction

Cuts up to 80-90% off monthly power bills by substituting costly grid units with free solar energy. Bi-directional net meter credits excess export.

Protected ROI

25-Year Long-Term Energy Security

Protects residential and commercial consumers from recurring DISCOM tariff hikes. High performance bifacial modules offer linear output up to 30 years.

PM Surya Ghar Ready

Direct Government Subsidy Support

Pre-qualified for central DBT subsidies up to ₹78,000 under PM Surya Ghar Yojana. Direct transfer to customer bank account post commissioning.

>98.6% Conversion Efficiency

Zero Battery Maintenance & Top Efficiency

Unlike off-grid systems, our on-grid topology has no recurring battery replacement expenses. High inverter efficiency (>98.6%) ensures maximum generation.

2. ENVIRONMENTAL IMPACT & GREEN ENERGY CONTRIBUTION

Environmental Metric	Estimated Offset / Contribution	Equivalent Impact
CO2 Emission Mitigation	~6.5 Metric Tons / Year	Reduces 162.5 MT carbon over 25 years
Equivalent Trees Planted	~310 Mature Trees	Natural carbon sequestration equivalent
Coal Consumption Conserved	~2900 kg Coal / Year	Replaces fossil-fuel powered thermal energy generation
Vehicular Miles Offset	~15,500 Passenger km / Year	Equivalent to removing 1 passenger car off the road

ACCELERATING CHHATTISGARH TOWARDS CLEAN ENERGY INDEPENDENCE

By adopting Rejoy Solar's high-efficiency 5 kW on-grid solar plant, you actively reduce state grid strain and thermal coal burning while creating measurable green savings. Your installation is fully compliant with CREDA and Ministry of New and Renewable Energy (MNRE) standards.

Rejoy Solar Power

3. HOW A SOLAR ON-GRID SYSTEM FUNCTIONS

STEP 1 — **Solar PV Array Absorption**
High-efficiency mono-crystalline/bifacial solar panels absorb solar irradiance on rooftop and generate direct current (DC) electricity seamlessly during daylight hours.

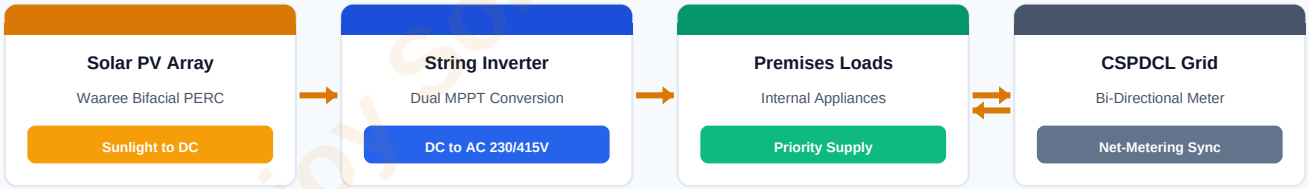
STEP 2 — **Synchronous Grid Inverter Conversion**
A state-of-the-art grid-tie string inverter transforms DC electricity into regulated 3-phase/single-phase alternating current (AC) matching utility grid frequency and voltage.

STEP 3 — **Priority Onsite Consumption**
Solar electricity first powers internal appliances, lighting, machinery, and air conditioning loads, drastically reducing instantaneous grid power draw.

STEP 4 — **Bi-Directional Net Metering Export**
Surplus solar energy generated beyond current load demand is automatically exported to the CSPDCL utility grid. The bi-directional net meter accurately logs export units.

STEP 5 — **Grid Offset & Nighttime Credit Settlement**
During nighttime or cloud cover, power is smoothly imported from grid. At the end of billing cycle, exported units offset imported units, leaving a nominal energy bill.

SYSTEM ARCHITECTURE & GRID INTERCONNECTION SCHEMATIC



** Real-time synchronization guarantees instant transition between onsite solar generation and utility grid power with zero interruption.*

Bi-Directional Power Flow: Day-time surplus exports to CSPDCL Grid; Night-time consumption draws from Grid with 1:1 billing offset.

4. WHY CHOOSE REJOY SOLAR POWER PVT. LTD.

Distinction	Rejoy Solar Engineering Edge	Client Value
CREDA Approved Grade-A	Officially empaneled EPC with Chhattisgarh State Renewable Energy Development Agency	100% compliant subsidy clearance
Tier-1 DCR PV Modules	Exclusively DCR Bifacial Dual-Glass Waaree modules with high temperature coefficient	Higher generation in Indian climate
Certified Civil HDG Structures	Wind rated up to 150 km/h with 80-120 micron hot-dip galvanization	Zero roof damage, rust-proof 25 yrs
Complete Liaisoning Service	End-to-end DISCOM approval, net meter testing, documentation and portal sanction	Hassle-free turnaround in 15-21 days
Live IoT App Monitoring	Cloud WiFi remote monitoring via smartphone application for daily kWh generation	Instant performance transparency

Rejoy Solar Power

5. COMPREHENSIVE TECHNICAL SPECIFICATIONS & SYSTEM COMPONENTS

System Subsystem	Component Details & Technical Standards	Compliance / Standard
Solar PV Modules	Waaree 545W DCR Bifacial Dual Glass Mono PERC Modules, 144 Half-cut cells, 21.5% Module Efficiency, IP68 Junction Box with bypass diodes	IEC 61215, IEC 61730, BIS Approved, ALMM Listed
Grid-Tied String Inverter	Waaree 5KW 3PH Grid Tie High Efficiency String Inverter, Dual MPPT, >98.6% European Efficiency, IP65 Waterproof Enclosure, Integrated WiFi Monitoring Dongle	IEC 61683, IEC 62109-1/2, IEEE 1547, VDE 0126
Module Mounting Structure	Elevated HDG (Hot-Dip Galvanized 80-120 micron) Rooftop Structure, Pre-GI C-channel purlins, SS304 Hardware, Wind Velocity Rated up to 150 km/h, 7-9 Ft clearance	IS 2062, IS 4759, IS 875 Part 3 (Wind Speed Certified)
DC Distribution Box (DCDB)	IP65 Polycarbonate Enclosure, 1000V DC Fuse Protection per string, Type-II Surge Protection Device (SPD), DC Isolator Switch	IEC 61439-1, DIN Rail Mounted
AC Distribution Box (ACDB)	IP65 Weatherproof Enclosure, 4-Pole / 2-Pole AC MCB / MCCB with Type-II AC Surge Arrester, Phase Indicator Lamps	IS / IEC 60947-2, IS 13947
Bi-Directional Net Meter	3-Phase / 1-Phase Bi-Directional Net Metering Kit approved by CSPDCL / DISCOM, Optical Port, DLMS Protocol	IS 13779, IS 14697, CEA Metering Regulations
Earthing & Surge Protection	Maintenance-free Chemical Earthing with Copper-bonded Electrodes & Backfill Compound (BFC), Copper/GI Flat Strips + Class-I Lightning Arrester	IS 3043 (Code of Practice for Earthing), IEEE 80
Solar DC & AC Cabling	1C x 4/6 sq.mm XLPO Electron Beam Cross-linked Solar DC Cable, UV Resistant, 2C/4C Armoured Aluminium/Copper AC Grid Cable	EN 50618 (TUV 2PfG 1169), IS 7098

MANUFACTURING QUALITY & MNRE DCR COMPLIANCE

All solar components supplied by Rejoy Solar Power Private Limited strictly comply with ALMM (Approved List of Models and Manufacturers) and Domestic Content Requirement (DCR) protocols required for government subsidy disbursement. Tier-1 solar modules are tested against potential induced degradation (PID) and salt mist corrosion.

Rejoy Solar Power

6. BILL OF MATERIALS (BOM) & EQUIPMENT SCHEDULE

S.No	Description of Goods	Make / Brand	Technical Specification	Qty	Unit	Rate (₹)	Amount
1	**DCR Bifacial Solar 545W Module**	Waaree	545Wp DCR Tier-1 Bifacial Dual Glass Mono PERC Modules	9	Nos	₹14,500	**₹1,30,500**
2	**Grid Tie Inverter 5KW 3PH (With WIFI Stick)**	Waaree	5 kW 3-Phase On-Grid String Inverter with remote WiFi monitoring	1	No	₹52,000	**₹52,000**
3	**ACDB - DCDB BOX 3PH - 1 in 1 out**	Waaree	IP65 Polycarbonate Enclosure, SPD Type-II, MCB/MCCB & Fuse Protection	1	Pair	₹9,500	**₹9,500**
4	**JSW/APL GI Module Mounting Structure Elevated**	JSW / APL Apollo	Hot-Dip Galvanized / Pre-GI elevated rooftop structure (80-120 micron)	1	Set	₹38,000	**₹38,000**
5	**Net Meter (CSPDCL) Genius/Equivalent**	Genius / Secure	3-Phase Bi-directional Net Meter approved by CSPDCL/DISCOM	1	No	₹8,500	**₹8,500**
6	**Earthing Kit**	Waaree	Chemical Earthing Electrodes with BFC compound & grounding strip	1	Set	₹7,500	**₹7,500**
7	**DC/AC/Earthing wires and cables**	Polycab / FlipCap / HPL	1C x 4/6 sq.mm Solar DC XLPO Cable, 4C Cu/Al AC armored cable	1	Lot	₹16,848	**₹16,848**
8	**PVC Pipes, MC4 Connectors, Tape, Joints, Cable Tray, Meter Board, Zip Tie, etc.**	As Standard	Balance of System: Conduit pipes, glands, lugs, fasteners, trays	1	Lot	₹6,000	**₹6,000**

TOTAL BILL OF MATERIALS (BOM) VALUE (EXCLUDING TAXES & SERVICES) **₹2,68,348**

SCOPE OF ENGINEERING, PROCUREMENT & COMMISSIONING (EPC)

- **3D Shadow Profiling & Topography**
  Comprehensive 3D shadow analysis, roof load assessment & civil inspection.
- **IS 875 Structural Design & Wind Safety**
  Complete structural calculation certified for wind velocity up to 150 km/h.
- **Tier-1 Procurement & Safe Logistics**
  Factory flash-tested DCR modules, inverters & balance of material delivered.
- **Elevated HDG Structure Erection**
  Hot-dip galvanized (80-120u) mechanical assembly & SS304 anti-theft fasteners.
- **DC/AC Cabling & Protection Pits**
  UV-resistant cross-linked solar DC cables, IP65 combiner boxes & chemical earthing.
- **CSPDCL / DISCOM Liaisoning**
  End-to-end DISCOM net metering sanction, CEIG approval & test documentation.
- **Plant Testing & IoT WiFi Telemetry**
  Insulation verification, inverter synchronization & smartphone cloud monitoring.
- **5-Year Comprehensive O&M Warranty**
  Preventive maintenance, scheduled module health checks & warranty support.

CLIENT SCOPE & PROJECT NOTES:

- Net Metering approval is subject to CSPDCL/DISCOM guidelines, transformer capacity, and sanctioned load.
- Client to provide adequate shadow-free roof terrace and auxiliary single-phase power/water during construction.
- All Balance of Material (BOM) items conform to IEC / BIS specifications with heavy-duty UV resistant conduits.
- Elevated structure provides 7-9 Ft walkability clearance with zero slab puncture using chemical anchoring.

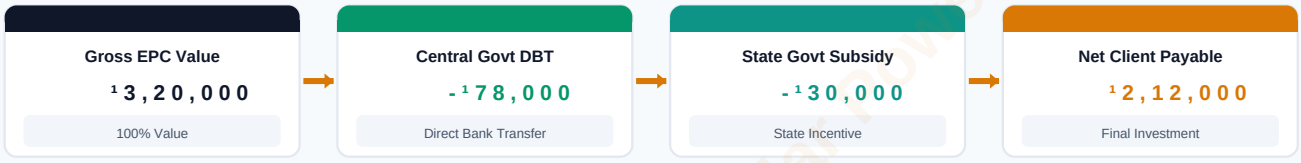
7. COMMERCIAL PRICE SCHEDULE & TAX STRUCTURE

Item	Description of Financial Head	Amount (INR)
1	Supply of Solar PV Modules, Inverter, Elevated Structure & Electrical BOS	₹2,68,348
2	Transportation, Freight, Handling & Transit Insurance to Site	₹5,500
3	Installation, Testing, Civil Grouting & Commissioning Services	₹20,000
4	Liaisoning, CSPDCL Documentation & Net-Metering Synchronization Charges	**Included in Turnkey EPC**
5	Gross Turnkey EPC Price before Discount & Taxes	₹2,93,848
6	Special Promotional / Scheme Discount	**Nil**
7	GST on Equipment Portion (70% of Base @ 5% GST)	₹10,285
8	GST on Services & Installation (30% of Base @ 18% GST)	₹15,868
9	**TOTAL PROJECT VALUE (INCLUSIVE OF APPLICABLE GST)**	**₹3,20,000**
10	Direct Central Government Subsidy (PM Surya Ghar Scheme)	-₹78,000
11	State Government Clean Energy Subsidy (If Applicable)	-₹30,000
12	**NET EFFECTIVE PROJECT INVESTMENT (POST SUBSIDY DISBURSEMENT)**	**₹2,12,000**

TOTAL PROJECT AMOUNT IN WORDS:

Three Lakh Twenty Thousand Rupees Only

COMMERCIAL INVESTMENT & GOVT SUBSIDY FINANCIAL WATERFALL



TOTAL GOVT SUBSIDY RELIEF: ₹1,08,000 (33.8% EFFECTIVE DISCOUNT ON GROSS VALUE)

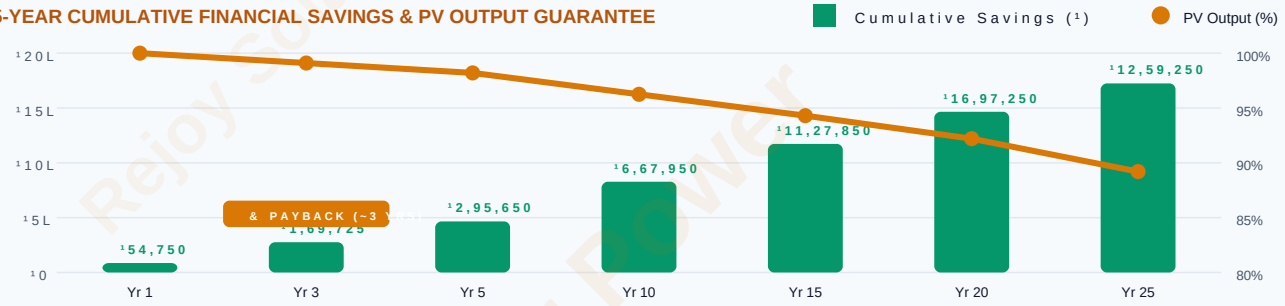
PM SURYA GHAR MUFT BIJLI YOJANA SUBSIDY DISBURSEMENT PROCESS

The central government DBT subsidy of ₹78,000 is directly transferred into the customer's bank account via the Nati
installation and DISCOM inspection. Rejoy Solar takes full responsibility for registration, document submission, portal uploading, and approval verification.

8. 25-YEAR ESTIMATED GENERATION & RETURN ON INVESTMENT (ROI)

ANNUAL GENERATION	1ST YR SAVINGS	PAYBACK PERIOD	25-YR NET SAVINGS
~7,300 kWh	₹ 5 4 , 7 5 0	~3.9 Years	₹ 1 2 , 5 9 , 2 5 0

25-YEAR CUMULATIVE FINANCIAL SAVINGS & PV OUTPUT GUARANTEE



Timeline	Performance Efficiency	Est. Power Output	Electricity Tariff	Annual Cost Savings	Cumulative Net Savings
Year 1	100.0%	7,300 kWh	INR 7.50 / unit	₹ 5 4 , 7 5 0	₹ 5 4 , 7 5 0
Year 2	99.4%	7,256 kWh	INR 7.73 / unit	₹ 5 6 , 0 8 9	₹ 1 , 1 0 , 8 3 9
Year 3	98.8%	7,213 kWh	INR 7.96 / unit	₹ 5 7 , 4 1 5	₹ 1 , 6 8 , 2 5 4
Year 4	98.2%	7,169 kWh	INR 8.20 / unit	₹ 5 8 , 7 8 6	₹ 2 , 2 7 , 0 4 0
Year 5	97.6%	7,126 kWh	INR 8.44 / unit	₹ 6 0 , 1 4 3	₹ 2 , 8 7 , 1 8 3
Year 10	94.7%	6,915 kWh	INR 9.79 / unit	₹ 6 7 , 6 9 8	₹ 6 , 2 5 , 6 7 3
Year 15	91.9%	6,710 kWh	INR 11.34 / unit	₹ 7 6 , 0 9 1	₹ 1 0 , 0 6 , 1 2 8
Year 20	89.2%	6,511 kWh	INR 13.15 / unit	₹ 8 5 , 6 2 0	₹ 1 4 , 3 4 , 2 2 8
Year 25	86.6%	6,318 kWh	INR 15.25 / unit	₹ 9 6 , 3 5 0	₹ 1 9 , 1 5 , 9 7 8

KEY ASSUMPTIONS IN FINANCIAL PROJECTIONS:

1. System yields ~4.2 daily sunshine peak hours in Chhattisgarh geographic climate zone.
2. Linear PV degradation factored at 0.6% annually as guaranteed by Tier-1 manufacturer warranty.
3. Base electricity tariff modeled at INR 7.50/unit with a modest conservative 3% annual grid tariff inflation.
4. Clean energy yields 100% self-consumed or compensated through CSPDCL bi-directional net metering protocol.

Rejoy Solar Power

9. ROOFTOP STRUCTURE DESIGN & SAFETY ENGINEERING

7 - 9 Ft Clear Headroom

Elevated High-Clearance Rooftop Structure

Engineered with 7 to 9 feet vertical clearance above terrace slab. Ensures 100% unrestricted rooftop terrace utilization for social events, clothes drying, and unobstructed water tank access.

80 - 120 Micron Zinc Coating

Hot-Dip Galvanized (HDG) Anti-Corrosion Coating

Structural steel components hot-dip galvanized with 80-120 micron zinc coating as per IS 4759 standards. Resists rust and chemical weathering for over 25 years without repainting.

IS 875 Part 3 Certified

150 km/h Wind Aerodynamic Rating

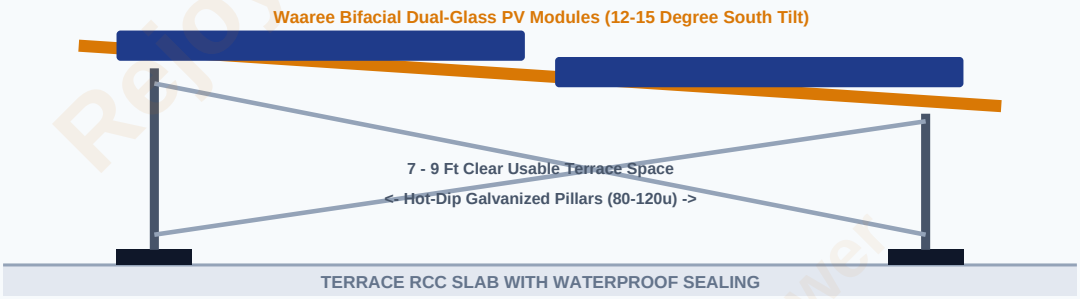
Structure designed and certified to endure severe cyclones and wind speeds up to 150 km/h in accordance with IS 875 Part 3 wind loading civil codes. Reinforced diagonal cross bracing.

100% Leak-Proof Grouting

Leak-Proof Chemical Fastening & Grouting

Heavy duty RCC base plates secured using chemical anchor fasteners and waterproofing seals. Zero puncturing or leakage into living areas beneath the terrace slab.

ELEVATED ROOFTOP MOUNTING STRUCTURE ENGINEERING SCHEMATIC



Rejoy Solar Power

10. MILESTONE PAYMENT SCHEDULE

Milestone Stage	Percentage	Amount (INR)	Trigger Event & Deliverables
10% Advance / Token	10%	₹ 32,000	Upon signing of the purchase order / contract
70% Material Procurement	70%	₹ 2,24,000	On delivery of PV modules and inverter at site
18% Installation Day	18%	₹ 57,600	On structure & module mounting completion
2% Grid Connection Completion	2%	₹ 6,400	On net-metering synchronization & final commissioning

11. GENERAL TERMS & SCOPE OF WORK MATRIX

Responsibility Head	Scope of EPC Contractor (Rejoy Solar)	Scope of Customer / Client
Rooftop Access & Civil	Structural design, elevated HDG fabrication, civil anchoring, module clamping, clean terrace handover.	Unobstructed roof access, shadow-free terrace area, water supply for cleaning modules.
Electrical & Grid Interconnection	Supply of DC/AC cables, ACDB/DCDB, inverter installation, chemical earthing, connection to main LT panel.	Adequate sanctioned load with DISCOM (CSPDCL), LT distribution board access.
Permits & Net Metering	Application filing, liaisoning with DISCOM, test report submission, inspection coordination, net meter installation.	Electricity consumer number (BP Number), ownership proof / electricity bills, KYC documents.
Project Timeline & Handover	System delivery in 1-2 weeks; installation completed in 3-5 days; net-meter synchronization per DISCOM schedule.	Timely release of payment milestone tranches as outlined in contract schedule.

Rejoy Solar Power

12. COMPREHENSIVE WARRANTY SCHEDULE

Component	Warranty Duration	Warranty Scope & Performance Commitment
Solar PV Modules	**10-12 Years Product / 25-30 Years Linear Output**	Guaranteed >90% power generation in first 10 years and >80% power output up to 25 years.
Solar Grid Inverter	**5 - 10 Years Comprehensive Replacement**	Free replacement / repair of faulty internal electronics, MPPT components, and WiFi modules.
Mounting Structure	**10 Years Structural & Galvanizing Warranty**	Protected against rust, corrosion, weld breakage, and mechanical wind failure.
EPC Workmanship	**5 Years Comprehensive O&M Support**	Complete system support including DC/AC connectors, earthing resistance, and periodic health checks.

13. OFFICIAL BANKING DETAILS FOR PAYMENTS

ACCOUNT NAME:	**REJOY SOLAR POWER PRIVATE LIMITED**	**IFSC CODE:**	**HDFC0001234 / ICIC0000123**
BANK NAME:	**ICICI Bank / HDFC Bank Limited**	**BRANCH:**	**Civic Centre, Bhilai, Chhattisgarh**
ACCOUNT NUMBER:	**50200088991234 (Current Account)**	**UPI ID:**	**rejoysolarpower@icici**

14. CONTRACT EXECUTION & CUSTOMER ACCEPTANCE

FOR REJOY SOLAR POWER PRIVATE LIMITED

Authorized Signatory
Designation: Commercial EPC Manager
Date of Issuance: 2026-09-21

CLIENT PURCHASE ORDER CONFIRMATION

I/We hereby accept the technical scope, pricing and milestone terms.

Shri Rameshwar Patel
Authorized Client Signatory / Property Owner
Date of Acceptance: _____

Rejoy Solar Power